Local rule activation and dealer capacity in a cellular automaton market*

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Abstract

Changes in investor decisions can destabilize a market, but a shift toward selling need not produce a price crash. We developed a market model in which traders decided whether to buy, hold, or sell from their neighbors' decisions or from changes in price. We compared markets with identical histories, then changed how selected traders in one market responded to their neighbors while leaving the comparison market unchanged. A majority decided to sell when enough traders received the new rule, even though the remaining traders kept their original rules. Setting the selected traders to Sell once without changing their rule for later decisions did not produce a majority, while several other changes to a single response did. A majority still formed when traders who normally used the price were kept in Hold, showing that responses to neighboring traders could sustain widespread selling on their own. Markets with more interacting traders shifted toward a majority over a narrower range of rule adoption, but the fraction needed continued to change with the time observed. The dealer's capacity to buy and sell affected how often a majority formed when prices influenced decisions, although greater capacity did not consistently make a majority more likely. Greater capacity did increase price declines in the comparison where a majority formed at every dealer capacity, even when we changed how strongly trades affected prices. Varying cash and share holdings explained this difference: traders needed resources to place orders, and the dealer needed resources to complete orders that other traders could not match. Similar Sell majorities could therefore accompany both small price declines and simulated crashes. The model connected investor decisions to market instability while showing how limited dealer resources could contain the price decline even when most traders continued deciding to sell.

*Repository: https://github.com/varnerlab/Wolfram-markets-abm-paper.git.

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1 Introduction

During the October 1987 market decline, portfolio insurance concentrated selling across participating traders [1, 2]. Forced sales by funds that had borrowed to invest spread losses across quantitative equity strategies in August 2007 [3]. Such episodes motivate a common modeling question: how can changes in some investors' decision rules destabilize a market and contribute to a crash? Agent-based models address collective behavior through imitation, strategy adaptation, and responses to a shared price [4–7]. Herding models describe how individual decisions produce collective activity [8–10], while lattice models restrict influence to neighboring traders [11–13]. These interactions can generate large fluctuations and periods of sustained volatility, two features of financial returns [14]. However, reproducing market-wide fluctuations does not by itself explain how the decisions of an interacting population change when some investors begin using a different rule to decide whether to buy, hold, or sell. Both the size of the changed group and the nature of the change matter. A temporary increase in selling and a persistent change in how traders respond to their neighbors need not produce the same collective outcome.

A cellular automaton represents investors as traders arranged on a square grid, with each trader deciding whether to buy, hold, or sell from the decisions of nearby traders. Each trader uses a lookup table to convert the sum of its four neighbors' numbered decisions into its next decision, a construction known as a totalistic rule [15–17]. Such models have been used to generate market fluctuations and changes in market behavior [12, 18–20]. One table entry can change while the other responses stay fixed. Asking a trader to sell once changes its current decision, whereas changing the rule changes how it responds whenever the same neighbor sum occurs again. The new rule remains in use even after the trader stops selling. Starting two markets from the same history, then changing the rule in only one, allows the effect of that change to be measured against continued use of the original rule [21, 22]. The question is how many traders must receive the new rule before at least half of the interior traders decide to sell, which we call a Sell majority, and whether this instability can develop without traders responding to price changes.

A market dominated by sellers and a market price crash are different outcomes. Their connection depends partly on the resources needed to trade. A trader needs shares to sell or cash to buy, and completing a trade requires someone willing and able to take the other side. Dealers can complete orders that other traders do not match, but their cash and share holdings limit how much they can buy and sell [23–27]. Price-impact models describe how completed trades change the price [28–31]. Agent-based markets have included limited trader resources, records of outstanding buy and sell orders, borrowing limits, and dealers with limited cash and shares [32–35]. In the grid model, traders can keep deciding to sell even when they cannot complete a sale, allowing instability in decisions to develop without a price crash. Limits on completed trades also change the price that informs later decisions. Dealer capacity can therefore affect both selling decisions and prices. An unlimited dealer can complete every order backed by trader resources, whereas a limited dealer may run out of cash or reach its limit on share holdings while many traders still want to sell.

In this study, we used a three-state cellular automaton market to determine how changes in investor decision rules caused market instability and price declines. We compared markets with the same history, then changed one response in the decision rule of selected interior traders. To test whether price changes were needed for a majority to form, we kept traders along the edge of the grid in Hold instead of letting them decide from the price. With these edge traders active and an unlimited dealer, changing the rule for 20% or 25% of interior traders produced a Sell majority in all 50 runs within 3,000 steps, whereas changing it for 10% or 15% produced none (Supplementary Table S2). Asking the same traders to sell once did not produce a majority, while several other changes to a single response in the rule did. Majorities also formed with edge traders kept in Hold, showing that decisions based on neighbors alone could sustain widespread selling. The fraction that needed the new rule nevertheless depended on grid size and observation time (Supplementary Table S3). When edge traders responded to price, greater dealer capacity did not consistently make a Sell majority more likely, but it did increase drawdown at 20% rule adoption. Changing trader cash and shares separately identified the resources that limited completed trades. Similar markets dominated by sellers accompanied median drawdowns from below 1% to above 25%, depending on how much selling could be completed (Fig. 2). Investor decisions could therefore create market instability without a price crash, because unmatched selling lowered the price only while the dealer could keep buying.

2 Methods

2.1 Three-state cellular automaton

We represented investors as traders on an $L \times L$ square grid, called a lattice. Traders inside the grid decided whether to buy, hold, or sell from their immediate neighbors' decisions. Selected traders along the edge used changes in price instead; keeping these edge traders in Hold allowed us to remove the influence of price on decisions. Buy ($s = 0$), Hold ($s = 1$), and Sell ($s = 2$) represented the resulting decisions, and the local rule specified which decision a trader made for a given neighbor configuration. Only the $(L-2)^2$ interior cells followed the cellular-automaton rule. All interior traders updated together from their four nearest neighbors, denoted by ∂i. Adding the numbered decisions of these neighbors gave the sum $c_i(t)$. A lookup table $\mathcal{R}$ assigned one next decision $a_j = \mathcal{R}(j)$ to each possible sum. A rule based only on this sum is called totalistic. The sum ranks Buy below Hold and Sell but does not distinguish all combinations of neighboring decisions. Two Holds and two Buys receive the same response as one Sell and three Buys. Changing one table entry therefore altered the response to a specified neighbor sum while preserving the other eight responses. Neighborhoods with the same sum remained equivalent under the altered rule. The local input and the Wolfram index R encoding the nine outputs were defined by the expressions [15, 16]:

$$c_i(t) = \sum_{j \in \partial i} s_j(t) \in \{0, 1, \ldots, 8\}, \tag{1}$$

$$R = \sum_{j=0}^{8} a_j 3^j. \tag{2}$$

Rule 19280, $(222110002)_3$, supplied the baseline decision rule. The activated Rule 19286, $(222110022)_3$, differed only at $c = 1$. A neighborhood containing three Buys and one Hold therefore returned Buy under the baseline and Sell under activation. Before each run, we selected a fraction f of interior sites uniformly without replacement as the target set for the later switch. The initial interior contained $\mathrm{round}(0.05(L-2)^2)$ Buy sites and the same number of Sell sites, with Hold elsewhere. Initial states were placed without replacement. A random-number seed identified each starting market and fixed the initial placements, selected traders, active edge traders, and their price-response parameters. It also fixed the priorities used when eligible orders exceeded dealer capacity. Once these choices were made, the market evolved deterministically. Delaying activation allowed the market to develop under the baseline rule before the response changed. At the intervention, each targeted trader retained its current decision and holdings but used the new rule to make subsequent decisions. Traders outside the target set kept their original decision rule throughout.

2.2 Orders, trader resources, and completed trades

A trader's decision generated a desired order, but carrying out that decision required sufficient cash or shares and a counterparty willing to take the other side. Every interior cell and active edge cell represented a trader with cash w_i, an integer number of shares q_i, and average purchase price $\bar{c}_i$. Buy requested the purchase of one share, Sell requested the sale of one share, and Hold requested no trade. An order was eligible, meaning backed by the required trader resources, if it could be paid for or supplied at the current price P_{t-1}. A buy required $w_i \geq P_{t-1}$, while a sell required $q_i \geq 1$. The main experiments disabled the optional cost-basis restriction, which permits sales only above average acquisition cost in the manner of the disposition effect [36, 37]. We excluded orders lacking the required cash or shares and recorded this exclusion as *agent suppression*. The trader's decision remained unchanged even though its order could not complete. A trader could therefore remain in Sell after exhausting its shares, or in Buy without enough cash to purchase a share. Its neighbors still responded to that state during the lattice update. We then matched the remaining orders with other traders or the dealer to determine which trades could complete (Algorithm S2).

We first matched buy orders with sell orders from other traders. With B_t eligible buys and S_t eligible sells, $\min(B_t, S_t)$ units on each side executed without using dealer resources. The dealer bought the shares that other traders did not buy, or sold the shares that other traders did not supply, within its cash and share limits. Capacity $h \geq 0$ specified net units per trader, giving total capacity $H = \mathrm{round}(hN_{\mathrm{tr}})$ for N_{tr} traders. A finite dealer began with cash $W_0 = P_0 H$ and inventory

$Q_0 = H$. It could not borrow and had to satisfy $0 \le Q_t \le 2H$. When buys exceeded sells, executed counts were:

$$n_t^S = S_t, \qquad n_t^B = S_t + \min(B_t - S_t, Q_{t-1}). \tag{3}$$

When sells exceeded buys, execution also depended on the dealer's remaining inventory space and ability to pay for shares:

$$n_t^B = B_t, \qquad n_t^S = B_t + \min\left(S_t - B_t, 2H - Q_{t-1}, \left\lfloor \frac{W_{t-1}}{P_{t-1}} \right\rfloor\right). \tag{4}$$

Eligible orders beyond these limits were *dealer-rejected*. When the dealer could not complete every remaining order, a reproducible priority score determined which traders could trade. The score depended on the seed, step, trader identity, and whether the order was a buy or sell. Including the step allowed priority to change over time while keeping it reproducible across paired runs. Completing trades at P_{t-1} updated trader holdings and changed dealer cash and shares by the corresponding amounts:

$$Q_t = Q_{t-1} + n_t^S - n_t^B, \qquad W_t = W_{t-1} + P_{t-1}(n_t^B - n_t^S). \tag{5}$$

These transfers conserved total cash and shares across traders and dealer, except when we intentionally changed trader cash or shares. The unlimited dealer, $h = \infty$, started with zero cash and shares but could borrow cash and sell borrowed shares to fill every eligible order. At $h = 0$, only opposing trader orders could clear. Both limits used the same clearing procedure (Algorithm S2).

2.3 How price changes influenced trader decisions

Traders along the edge of the grid brought price information into neighboring traders' decisions. These edge traders compared the current price with recent prices, while the interior traders used their neighbors' decisions. The difference between completed trader buys and sells determined the price change. We normalized the difference between filled buys and sells by the fixed trader count N_{tr} to obtain imbalance ϕ_t. Positive imbalance raised the price, negative imbalance lowered it, and balanced execution left it unchanged. The exponential update used price-impact coefficient $\lambda = 0.02$ and initial price $P_0 = 100$. Active edge traders compared each new price with a reference price μ_t, updated from recent prices, and a measure σ_t of recent price departures from that reference. Each trader had a fixed multiplier $k_j \sim U[k(1-\delta), k(1+\delta)]$, with $k = 1$ and $\delta = 0.5$. The price range in which this trader chose Hold extended from $\mu_t - k_j\sigma_t$ to $\mu_t + k_j\sigma_t$. The trader chose Buy below this range, Sell above it, and Hold inside it. One quarter of perimeter cells were active; the others stayed at Hold. These decisions influenced adjacent interior traders, allowing price changes from completed trades to affect later decisions inside the grid. Imbalance, price, and the reference price and measure of recent price variation followed these updates, with smoothing parameter $\alpha = 1/60$:

$$\phi_t = \frac{n_t^B - n_t^S}{N_{\mathrm{tr}}}, \tag{6}$$

$$P_t = P_{t-1}\exp(\lambda\phi_t), \tag{7}$$

$$\mu_t = \mu_{t-1} + \alpha(P_t - \mu_{t-1}), \qquad \sigma_t = \sigma_{t-1} + \alpha(|P_t - \mu_t| - \sigma_{t-1}). \tag{8}$$

Orders came from the current trader states, while the resulting price influenced the next interior update (Algorithm S1). At each step, rule activation and any financial intervention preceded eligibility tests, clearing, and settlement. Executed orders then generated the new price. Boundary traders responded to the price before the interior cells updated synchronously. The complete step followed the sequence:

$$\text{lattice state} \to \text{orders} \to \text{dealer fills} \to P_t \to \text{boundary state} \to \text{lattice state}.$$

To prevent price changes from affecting decisions, we set every active edge trader to Hold before each interior update. We call this condition *held boundary*; *endogenous boundary* denotes the condition in which edge traders responded to the price generated by trades. The comparison kept the same initial grid, selected traders, active edge locations, price-response parameters, trader holdings, and dealer. Orders still passed through the same eligibility, clearing, and price rules. Cash, shares, and price could still change, but these changes could no longer affect decisions inside the grid. Interior traders responded only to neighboring decisions, allowing us to determine whether a Sell majority could form without information about price changes (Algorithms S1 and S2).

2.4 Paired rule-switch experiments

We compared the consequences of changing a decision rule with continued use of the original rule in paired markets that shared a common history. Both paths used Rule 19280 at every interior site for $T_{\mathrm{pre}} = 1{,}000$ baseline steps. At the start of step $T_{\mathrm{pre}} + 1$, one path activated Rule 19286 at the preselected fraction f. The matched control left those sites on Rule 19280. Neither path reset the lattice, price, edge traders' price-response ranges, trader holdings, or dealer. Their common state $X_{T_{\mathrm{pre}}}$ also generated the first price after activation, so the earliest possible difference occurred in the following interior update. We checked the shared pre-switch histories and first post-switch price bit for bit in every pair. Each switched path thus began with the same trader resources and neighbor configurations as its control. The four main sets of simulations used 50 seeds and 3,000 steps after the rule change, with initial trader cash of 1,000 and ten shares each. Additional comparisons tested other rules, a one-time instruction to sell, fixed widths for the edge traders' Hold ranges, different strengths of the price response to trades, and longer observation periods. The supplementary protocols specify which part of the intervention or feedback was changed in each control (Appendix A.5).

We varied the fraction of traders given the new rule, grid size, and financial resources to determine when a Sell majority formed and how many sales could complete (Supplementary Table S1). The selected switch experiment used $L = 64$, an unlimited dealer, and an endogenous boundary at four activation fractions. We then tested more fractions at eight dealer capacities, both with edge traders responding to price and with those traders kept in Hold. The finite-size campaign used $L \in \{32, 48, 64, 96, 128\}$ under a held boundary with $h = \infty$ and endogenous boundaries with $h = 1$ or ∞. Because capacity was defined per trader, absolute dealer resources increased with lattice size. The financial campaign multiplied current trader cash and shares independently by $g_w, g_q \in \{0.1, 1, 10\}$ at activation, rounding shares to integers. Its nine interventions shared the same pre-intervention path within each $(f, h, \text{boundary}, \text{seed})$ cell. Under the held boundary, they also had to retain identical lattice trajectories afterward because neither cash nor shares entered the interior update directly. A discrepancy would therefore indicate an implementation error. The appendix gives the full parameter grids and the checks used to verify these paired and held-boundary relationships.

2.5 Market outcomes, statistical analysis, and selection of examples

We used the strictly interior Sell fraction $m(t)$ to measure the shift toward selling. Larger values indicated that more interior traders occupied Sell, and we treated the emergence of a Sell majority after the rule change as a sign of market instability. We recorded a Sell majority when $m(t) \geq 0.5$ at least once after the rule change. The onset time was the first step that reached this value, and majority probability, also called transition probability, was the fraction of runs that reached it. We summarized onset times only for runs that reached a majority and averaged the Sell fraction over the final 500 steps to describe each market near the end of the observation period. To measure how much this final mean $\bar{m}$ differed between runs, we multiplied its variance by the number of interior traders, $(L-2)^2$. Angle brackets denote averages across seeds. The resulting quantity, called susceptibility χ_L, is used to compare systems of different sizes [38, 39]:

$$\chi_L = (L-2)^2 \left[\langle \bar{m}^2 \rangle - \langle \bar{m} \rangle^2 \right]. \tag{9}$$

At a fixed grid size, higher susceptibility indicated greater variation in the final mean Sell fraction between starting markets. Linear interpolation gave $f_q(L)$, the activation fraction at which the transition probability reached $q/100$. The midpoint was $f_{50}(L)$, and the width $f_{75}(L) - f_{25}(L)$ measured how much additional activation separated probabilities of 0.25 and 0.75. A smaller width therefore indicated a sharper response. Interpolation used the first upward crossing without forcing the sampled curve to be monotone. Binomial proportions used 95% Wilson intervals [40]; continuous summaries used interquartile ranges unless otherwise specified.

A high Sell fraction need not produce a large price decline if the requested orders cannot execute. We measured the decline by maximum drawdown, the largest proportional fall from a running price peak during the post-activation window. We used price crash to describe a large simulated decline and reported its magnitude through drawdown, without defining a cutoff for classifying runs as crashes. We recorded unfilled orders at the stage where resources prevented execution. Agent suppression counted desired orders excluded by insufficient trader cash or shares. Dealer rejection counted eligible orders left unfilled after matching because the dealer lacked cash or inventory capacity. In the production campaigns, the reported rejection fraction used all eligible orders over the

full run, including the baseline steps, as its denominator, whereas the supplementary controls used post-activation orders only. Higher rejection fractions meant the dealer prevented more orders backed by trader resources from completing. Financial summaries used medians and interquartile ranges across seeds. Drawdown measured the price decline caused by completed trades, while suppression and rejection identified why desired trades could not complete.

Testing too few fractions of traders with the new rule could miss the range over which majority probability changed. We refined each curve using a common resolution criterion. Intervals intersecting the empirical 5–95% response were bisected until every size and boundary/capacity condition had at least five fractions strictly inside that probability range. Each added fraction used the original 50 seeds. We then resampled whole seeds across all fractions 2,000 times to estimate uncertainty in the crossings, width, and sampled susceptibility maximum. These percentile intervals preserved dependence between fractions and included reselection of the susceptibility peak. We also fitted a logistic curve, which required majority probability $p_L(f)$ to increase smoothly with the fraction f. The fitted parameters a_L and b_L controlled its position and slope:

$$p_L(f) = \{1 + \exp[-a_L - b_L(f - 0.18)/0.01]\}^{-1}, \qquad b_L > 0.$$

We fitted this curve by binomial likelihood and used the same seed blocks to bootstrap its midpoint and width. Fitted standard errors were bootstrap standard deviations. Both analyses measured uncertainty for the tested fractions and observation period. Neither included error from interpolation, the choice of additional fractions, or observing the market for longer.

We selected examples of traders' decisions across the grid using the measured time to a majority or final Sell fraction to choose each run. Selection preceded inspection of the images. The activation-fraction panels included a run without a majority at $f = 0.15$, both outcomes at $f = 0.175$, and runs with a majority at $f = 0.185$ and 0.20 under $h = \infty$. Among runs that reached a majority, we selected the run nearest the group's median onset time. Among runs without a majority, we selected the run nearest the group's median final Sell fraction. The dealer panels required a single seed that transitioned at $h = 0.1$, 1, and ∞ for $f = 0.20$. Among eligible seeds, selection minimized the summed distance from the three capacity-specific median onsets. Both panel sets showed snapshots at activation and at offsets 500, 1,500, and 3,000, allowing the spatial trajectories to be compared at common elapsed times. The examples therefore had typical onset times for their groups with a majority and typical final Sell fractions for their groups without one. Individual runs illustrated how groups of sellers developed, while probabilities across runs measured how often a majority occurred.

3 Results

To explore how investor decisions could destabilize a market, we changed how some traders responded to their neighbors and found that a majority entered the Sell state only when enough traders adopted the new decision rule (Fig. 1). We compared each market with an otherwise identical market where traders continued to use their original decision rules. The new rule, Rule 19286, changed one decision from Buy to Sell, and we assigned it to four different fractions of traders inside the grid. We used a 64×64 grid with an unlimited dealer and traders along its edge who decided whether to buy, hold, or sell from changes in the price (Supplementary Table S1). Each pair of markets had the same 1,000-step history and the same price immediately after we assigned the new rule. They first differed when the selected traders next decided how to act with three neighbors buying and one holding. All 50 markets that kept the original rule remained below a Sell majority for the next 3,000 steps. The same was true when we changed the rule for a fraction $f = 0.10$ or 0.15 of interior traders, whereas changing it for $f = 0.20$ or 0.25 produced a majority in every run. The median time to reach a majority fell from 1,044.5 to 311.5 steps between the two larger fractions (Fig. 1B; Supplementary Table S2). In the representative $f = 0.20$ pair, groups of sellers expanded across the grid after the rule change while decisions in the comparison market remained mixed (Fig. 1C). We also tested whether a temporary decision to sell had the same effect as changing the rule for later decisions. Setting the same selected traders to Sell once, while keeping their original rule, produced no majority at $f = 0.15$, 0.20, or 0.25, whether or not traders at the edge responded to price (Supplementary Table S5). Of all 18 ways to change a single response in the original rule, Rule 19280, six produced a majority at $f = 0.20$ and the other twelve did not (Supplementary Table S6). Within these tests, the shift to a market dominated by sellers depended on how many traders changed their decision rule, which decision changed, and whether the new rule remained in use.

Traders inside the grid decided from their neighbors' actions, while traders along its edge used the price and could therefore pass the effects of a price change to their neighbors. To determine whether a Sell majority could form without that influence, we changed the decision rule for selected interior traders while keeping all traders at the edge in Hold, and found that a majority still formed (Fig. 2). With the edge traders unable to buy or sell, the fraction of runs reaching a majority rose from 0.34 at $f = 0.175$ to 0.70 at 0.185 and 1.00 at 0.20 (Fig. 2A). Changing dealer capacity left the entire sequence of interior decisions unchanged for each starting market and fraction f, even though completed trades, prices, and dealer holdings differed. Prices could still change after trades, but no trader could use those prices to change a decision. Responses to neighboring traders were therefore sufficient to sustain widespread selling. Allowing traders at the edge to respond to price again changed how often a majority formed only among fractions where some runs reached a majority and others did not (Fig. 2A,B). With these traders active and an unlimited dealer, 14 of 50 runs reached a majority at $f = 0.175$, 43 did so at 0.185, and all 50 did so at every tested fraction from 0.195 upward. Across all dealer capacities with these traders active, no run reached a majority at $f \leq 0.15$ and every run did so at $f \geq 0.20$. Responses to price thus changed how often a majority formed but were not required for it to develop. We next asked how this instability depended on the number of interacting traders and the time available, and when it was accompanied by a large price decline.

The fraction of traders needing the new rule could depend on how many traders interacted and how much time their decisions had to influence one another. We changed the rule at five grid sizes while keeping edge traders in Hold, and found that larger markets needed a smaller fraction of traders to receive the new rule for a majority to form (Fig. 3). We tested additional fractions near the rise in majority probability to measure its position and width. After 3,000 steps, the fraction at which half the runs reached a majority, f_{50}, fell from 0.216 at $L = 32$ to 0.166 at $L = 128$. The additional fraction needed to raise majority probability from 0.25 to 0.75 decreased from 0.023 [0.018, 0.027] to 0.008 [0.004, 0.010], with brackets giving 95% confidence intervals obtained by resampling the starting markets (Fig. 3A,C; Supplementary Table S3). We also measured variation between runs by multiplying the variance of the final mean Sell fraction by the number of interior traders. The maximum value of this measure, susceptibility, rose from 21.2 to 241.2 over the same sizes (Fig. 3B,D). Fitting smooth curves that required probability to increase with rule adoption placed each midpoint within 0.004 of the estimate from interpolation between tested fractions, although the fitted widths were narrower when edge traders stayed in Hold (Supplementary Table S10). Allowing more time also changed the result: extending the $L = 64$ runs from 3,000 to 9,000 steps lowered the fraction needed for half the runs to reach a majority from about 0.178 to 0.167. The larger of the two endpoint grids still needed a lower fraction when we adjusted observation time with grid size or changed the Sell fraction required to count a transition, but the estimated midpoints kept moving in longer runs (Supplementary Table S9). The sharper rise in majority probability on larger grids therefore described the markets over the time observed; it did not establish a threshold that would remain fixed as observation time increased.

The frequency of a Sell majority across runs did not reveal how traders' decisions developed within an individual market. To determine whether selling spread through neighboring traders or appeared across the market at once, we followed decisions at successive times and found that each displayed majority developed as groups of sellers grew and joined together (Fig. 4). We selected these runs from their summary measurements before inspecting any images, choosing examples with typical times to a majority or typical final Sell fractions for their outcome group. At $f = 0.15$, the run without a majority retained a mixture of decisions across the grid for all 3,000 steps. The two $f = 0.175$ runs followed different paths: one formed a group of sellers but stayed below one-half, while the other developed a large connected group and reached a majority after 2,186 steps. The $f = 0.185$ run reached a majority after 1,779 steps through several connected groups, and the $f = 0.20$ run did so after 1,030 steps and ended with one dominant group of sellers (Fig. 4A). Price declined as the fraction of sellers rose in each run that reached a majority. Maximum drawdowns were 17.6%, 23.7%, and 25.6% at $f = 0.175$, 0.185, and 0.20, compared with 5.3% and 9.9% for the two runs without a majority (Fig. 4B,C). The number, shape, and timing of the groups varied across starting markets. The $f = 0.175$ run that never reached a majority also formed a visible group of sellers, so the appearance of such a group did not guarantee a majority within the time observed. In each displayed transition, local decisions formed growing groups of sellers rather than a single market-wide decision to sell. The probability measured across runs described how often those interacting decisions produced a majority.

A decision to sell could influence neighboring traders without finding a buyer, so a market dominated by sellers did not necessarily undergo a price crash. We varied the dealer's ability to buy and sell across eight capacities while allowing traders at the edge to respond to price, and found that greater capacity increased the price decline but did not consistently increase the probability of a Sell majority (Fig. 2). At $f = 0.175$, the probability was 0.66 with capacity $h = 0.1$ and 0.28 with unlimited capacity, a difference of 38 percentage points in favor of the smaller dealer (Fig. 2A,B). The edge traders normally adjusted how large a departure from their reference price would trigger a decision to buy or sell. Keeping that required departure fixed at its value before the rule change changed the probability difference from 0.38 to -0.02. This linked the smaller dealer's advantage to traders adjusting their responses to price, although other increases and decreases in probability across capacities remained (Supplementary Table S7). At $f = 0.20$, every run reached a majority at every capacity, but a larger dealer allowed more orders backed by trader cash or shares to complete. Median maximum drawdown rose from zero at $h = 0$ to 0.40% at $h = 0.1$, 3.92% at $h = 1$, and 25.3% with unlimited capacity, while the median fraction of these orders rejected by the dealer fell from 0.449 to zero (Fig. 2C,D). Without dealer capacity, traders could only buy from or sell to one another, so equal numbers of completed buys and sells left the price unchanged. Capacity $h = 10$ rejected no orders and matched unlimited capacity for the resources and duration tested. Larger dealers still produced larger drawdowns when we varied how strongly completed trades changed the price over a tenfold range, although drawdown magnitudes changed by a factor of seven to ten (Supplementary Table S8). Runs from the same starting market also separated selling decisions from price declines. At $f = 0.20$, the market initialized with seed 25 reached a majority at $h = 0.1$, 1, and ∞ within 88 steps of one another. The mean Sell fraction over the final 500 steps lay between 0.81 and 0.87 in these three runs and in the comparison with edge traders kept in Hold, yet maximum drawdown rose from 0.40% at $h = 0.1$ to 3.92% at $h = 1$ and 25.8% with unlimited capacity (Fig. 5). Similar instability in decisions thus accompanied both a small decline and a simulated price crash. Dealer capacity limited the price decline after most interior traders had decided to sell, while the resulting price influenced later decisions.

Traders needed shares to carry out a decision to sell and cash to carry out a decision to buy, even when their decision rules called for those actions. To distinguish the effects of trader resources from the dealer's ability to complete their orders, we multiplied trader cash and shares independently by 0.1, 1, or 10 when we changed the rule, and found that holdings changed majority probability only when traders could respond to price (Fig. 6). We tested these changes at three dealer capacities, both with traders at the edge responding to price and with those traders kept in Hold. At $f = 0.18$, all nine combinations of cash and shares produced identical interior decisions when edge traders stayed in Hold, with a majority probability of 0.64 across dealer capacities. When edge traders responded to price, probabilities ranged from 0.50 to 0.94, and the other tested fractions showed the same dependence on whether these traders were active (Fig. 6A; Supplementary Table S4). Holdings determined whether traders could carry out their decisions, but changed later decisions only through their effect on price. With the original cash holdings and an unlimited dealer, increasing the share multiplier from 0.1 to 1 to 10 raised median drawdown from 2.87% to 20.1% to 83.5%, because every sell order backed by a share completed and the dealer bought all shares that other traders did not buy (Fig. 6B). Limited dealer capacity capped that increase: the original and tenfold share holdings both produced 3.92% at $h = 1$ and 0.399% at $h = 0.1$, because the dealer rejected sell orders beyond its capacity (Fig. 6C). Majority probability did not follow drawdown (Fig. 6D). At $h = 1$ with the original share holdings, tenfold cash reduced median drawdown from 3.92% to 0.089% while raising majority probability from 0.78 to 0.94. The extra cash allowed more traders to buy, so they bought shares that would otherwise have needed the dealer as a buyer, while the changed price also influenced decisions at the edge. Trader and dealer resources therefore determined whether a market dominated by sellers experienced a limited decline or a large price crash. The decision to sell and the resources needed to complete the sale contributed separately to the market outcome.

4 Discussion

Changes in investor decision rules generated market instability, while available trading resources determined whether the shift toward selling accompanied a small decline or a price crash. When enough traders received a new rule for responding to their neighbors, a majority entered Sell even though the remaining traders kept their original rules. The new rule remained in place when a selected trader stopped selling, so the changed response applied whenever the same combination of

neighboring decisions occurred again. Asking those traders to sell once changed only their current decision and produced no majority in the 50 runs tested for each condition. Keeping Rule 19286 in use produced a majority in every run at the two larger fractions, and several other changes to a single response also produced majorities, although most rules tested did not. Widespread selling therefore depended on which response changed as well as on how many traders received the new rule. Comparing markets with identical histories isolated the effect of the rule change. Repeating that comparison across starting markets then measured how often a majority formed at each fraction of traders given the new rule.

Market instability could develop through neighboring traders' decisions without traders responding to a price change. Keeping traders at the edge in Hold removed the influence of price on decisions inside the grid, yet Sell majorities still developed. In the displayed runs that reached a majority, groups of sellers formed, enlarged, and joined together. Local decisions thus produced growing spatial patterns, as occurs in other models of interacting sites away from equilibrium [41, 42]. The dependence on the fraction of traders given the new rule resembled models in which the size of an initially changed group determines whether a change spreads across a population [43–46]. The fraction needed for a majority depended on both the number of interacting traders and the time allowed. On larger grids, the probability rose from low to high over a narrower range of rule adoption, but allowing more time continued to lower the fraction at which half the runs reached a majority. We therefore did not identify a threshold that would remain fixed in larger markets or longer runs. Establishing such a limit would require the estimates to stop changing with observation time and a mathematical model for their dependence on grid size [38, 39]. The growing groups suggested an analogy with droplets forming in a phase change, but their appearance did not establish the physical mechanism of classical nucleation. We did not identify an energy barrier, a formation rate, surface tension, or a critical group size that separated growth from decay.

The link between market instability and a price crash depended on whether investors could carry out their decisions. Traders first needed cash to buy or shares to sell, after which buy and sell orders were matched between traders. The dealer completed remaining orders only while its cash and share limits allowed. These limits connect the model to accounts of how dealer holdings affect trading [23–25, 35]. Only the difference between completed trader buys and sells changed the price [28, 30, 31]. Without dealer capacity, equal numbers of completed buys and sells left the price fixed, whereas unlimited capacity allowed every order backed by trader resources to complete. Runs from the same illustrated starting market consequently reached a Sell majority with either a 0.40% or a 25.8% drawdown (Fig. 5). At 18% rule adoption with the original cash holdings, tenfold trader shares produced an 83.5% median drawdown against the unlimited dealer, but limited capacity held the corresponding drawdowns to 3.92% or 0.399% (Fig. 6). More shares allowed traders to place more sell orders, but selling beyond what other traders bought lowered the price only while the dealer could continue buying. A market dominated by sellers therefore did not determine the severity of the crash on its own, because the price fell only as far as someone was able to buy.

Completed trades changed the price, which could then influence later decisions through the traders at the edge of the grid. At $f = 0.175$, capacity $h = 0.1$ raised majority probability by 38 percentage points relative to the unlimited dealer (Fig. 2). Edge traders compared the price with a moving reference and adjusted the size of the departure that would trigger buying or selling. Keeping this required departure fixed at its value before the rule change removed the smaller dealer's advantage in the estimated probabilities, linking the advantage to traders adjusting their responses to price (Supplementary Table S7). The reference price still changed, as did trader and dealer holdings, and the fixed departures still differed across starting markets and capacities. Increasing capacity also continued to produce both increases and decreases in majority probability, so this adjustment did not explain the entire effect of dealer capacity. Changing how strongly trades moved the price gave another limit on the interpretation: larger dealers still produced larger drawdowns across a tenfold range of price-impact coefficients, but drawdown magnitudes changed by a factor of seven to ten (Supplementary Table S8). Limiting completed trades could therefore reduce drawdown while the resulting price changes made a Sell majority more likely. The two outcomes measured how many traders decided to sell and how far completed trades lowered the price. A Sell majority alone was insufficient to identify a price crash.

Several modeling choices limit how far these conclusions extend. Traders followed assigned decision rules rather than learning a strategy or choosing one to maximize returns. They updated together on a regular grid using one original rule, and different combinations of neighboring decisions could have

the same sum and therefore produce the same response. Testing changes to this lookup table did not cover all possible ways to respond to neighboring traders. Each order requested one share at a common price, and the dealer had no strategy for setting prices or managing risk. It could replenish cash only by selling shares and replenish shares only by buying them. The model included resource limits but no record of outstanding orders at different prices, and its price response and time steps were not calibrated to an observed market. Its simulated crashes therefore did not predict the timing or magnitude of actual market crashes. Ten runs with the original decision rule in each condition, with edge traders either responding to price or kept in Hold, remained below a Sell majority for 100,000 additional steps (Appendix A.5). This extended the observed persistence of mixed decisions without establishing that it would continue indefinitely, and neither the 1,000-step initial period nor the final 500-step average established that the distribution of market states had stopped changing. Updating traders at different times, using other decision rules, or adding an order book would allow the same question to be examined under different assumptions: how investor decisions destabilize a market, and what determines whether that instability accompanies a crash.

5 Conclusion

Changes in investor decision rules destabilized the modeled market when enough traders received a new response to their neighbors. A Sell majority emerged while otherwise identical markets retained mixed decisions under the original rule. Asking the same traders to sell once did not produce these majorities, while several other changes to a single response did. The fraction of traders that needed the new rule depended on grid size and observation time. Keeping traders at the edge in Hold showed that decisions based on neighbors alone could sustain widespread selling without information about price changes. Completing those sales required shares, buyers with cash, and dealer resources for orders that other traders could not match. When edge traders responded to price, greater dealer capacity increased drawdown in the comparison where every capacity produced a Sell majority even though a smaller dealer could make a majority more likely at some lower fractions. Market instability and a price crash were therefore separate outcomes of investor decision-making in this model. Exhausted dealer capacity limited the price decline even while many traders continued deciding to sell.

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Tables

Table S1: Design of the four main sets of simulations. "Endogenous" means edge traders responded to price; "held" means they stayed in Hold. "Endogenous/held" indicates that both conditions were tested.

Campaign	Lattice sizes	Dealer h	Boundary	Seeds	Simulations
Rule change	64	∞	endogenous	50	400
Dealer capacity	64	0–10, ∞	endogenous/held	50	9,600
Grid size	32–128	$1, \infty$	three conditions	50	9,750
Cash and shares	64	$0.1, 1, \infty$	endogenous/held	50	10,800

Table S2: Comparison of changed and unchanged decision rules on a 64×64 grid with an unlimited dealer and edge traders responding to price. Transition probability is the fraction of runs in which at least half of the interior traders decided to sell after the rule change. Median onset is the median time to reach this condition among runs that did so; it is undefined where no run reached a majority.

		Transition probability		
Fraction f	Seeds	No-switch	Switched	Median onset (steps)
0.10	50	0.00	0.00	Undefined
0.15	50	0.00	0.00	Undefined
0.20	50	0.00	1.00	1,044.5
0.25	50	0.00	1.00	311.5

Table S3: Effect of grid size over 50 starting markets and 3,000 steps after the rule change. Brackets give 95% confidence intervals (CI) from seed resampling. n_{int} counts fractions with $0.05 < \hat{p} < 0.95$. Crossings use linear interpolation. The susceptibility maximum is over the sampled grid, and its interval includes reselection of that maximum within each bootstrap replicate. Intervals condition on the refined grid.

Boundary	h	L	n_{int}	f_{50} [95% CI]	Width [95% CI]	Peak χ_L [95% CI]
Endog.	1	32	8	0.210 [0.207, 0.218]	0.024 [0.018, 0.031]	23.6 [19.4, 30.0]
Endog.	1	48	5	0.186 [0.182, 0.188]	0.016 [0.010, 0.020]	41.3 [32.5, 56.2]
Endog.	1	64	8	0.178 [0.172, 0.178]	0.009 [0.007, 0.011]	70.1 [56.1, 91.8]
Endog.	1	96	6	0.166 [0.164, 0.168]	0.010 [0.007, 0.012]	126.5 [104.2, 163.7]
Endog.	1	128	5	0.165 [0.164, 0.166]	0.006 [0.005, 0.007]	191.4 [154.7, 261.0]
Endog.	∞	32	8	0.212 [0.208, 0.215]	0.023 [0.011, 0.030]	23.9 [18.5, 31.7]
Endog.	∞	48	6	0.188 [0.179, 0.192]	0.019 [0.016, 0.021]	52.0 [42.7, 67.5]
Endog.	∞	64	5	0.179 [0.177, 0.181]	0.009 [0.006, 0.014]	90.3 [78.2, 121.0]
Endog.	∞	96	5	0.168 [0.166, 0.169]	0.010 [0.005, 0.012]	173.4 [136.1, 230.6]
Endog.	∞	128	7	0.165 [0.164, 0.167]	0.007 [0.005, 0.008]	262.5 [226.5, 338.3]
Held	∞	32	5	0.216 [0.211, 0.224]	0.023 [0.018, 0.027]	21.2 [16.5, 27.2]
Held	∞	48	5	0.193 [0.184, 0.195]	0.019 [0.016, 0.022]	48.6 [40.3, 65.3]
Held	∞	64	5	0.178 [0.176, 0.180]	0.015 [0.008, 0.019]	110.5 [78.5, 138.5]
Held	∞	96	5	0.168 [0.167, 0.170]	0.009 [0.006, 0.012]	168.7 [137.6, 221.0]
Held	∞	128	7	0.166 [0.165, 0.167]	0.008 [0.004, 0.010]	241.2 [202.6, 326.0]

Table S4: Probability of a Sell majority after changes in trader cash and shares, over 50 starting markets. Each fraction includes 27 combinations of cash, shares, and dealer capacity. Keeping edge traders in Hold gave the same probability for all combinations; allowing them to respond to price gave the stated range.

Fraction f	Held boundary	Endogenous range
0.170	0.22	0.08–0.52
0.175	0.34	0.24–0.82
0.180	0.64	0.50–0.94
0.185	0.70	0.64–0.98

Table S5: Asking selected traders to sell once (Sell pulse) versus keeping Rule 19286 in use. Both comparisons used the same traders and 3,000 steps after the change. Entries are majority probabilities with 95% Wilson intervals; the dealer was unlimited under both boundary conditions.

Boundary	f	Seeds	Sell pulse	Persistent rule
Endogenous	0.15	50	0.00 [0.00, 0.07]	0.00 [0.00, 0.07]
Endogenous	0.20	50	0.00 [0.00, 0.07]	1.00 [0.93, 1.00]
Endogenous	0.25	50	0.00 [0.00, 0.07]	1.00 [0.93, 1.00]
Held	0.15	50	0.00 [0.00, 0.07]	0.00 [0.00, 0.07]
Held	0.20	50	0.00 [0.00, 0.07]	1.00 [0.93, 1.00]
Held	0.25	50	0.00 [0.00, 0.07]	1.00 [0.93, 1.00]

Table S6: All changes to one response in Rule 19280. Each probability pair is edge traders kept in Hold / responding to price over 20 starting markets; the changed rule remained in use. The output labels are Buy $= 0$, Hold $= 1$, Sell $= 2$.

Sum c	Output change	Rule	$f = 0.15$	$f = 0.20$	$f = 0.25$
0	$2 \to 0$	19278	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
0	$2 \to 1$	19279	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
1	$0 \to 1$	19283	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
1	$0 \to 2$	19286	0.00 / 0.00	1.00 / 1.00	1.00 / 1.00
2	$0 \to 1$	19289	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
2	$0 \to 2$	19298	0.95 / 1.00	1.00 / 1.00	1.00 / 1.00
3	$0 \to 1$	19307	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00
3	$0 \to 2$	19334	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00
4	$1 \to 0$	19199	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
4	$1 \to 2$	19361	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00
5	$1 \to 0$	19037	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
5	$1 \to 2$	19523	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00
6	$2 \to 0$	17822	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
6	$2 \to 1$	18551	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
7	$2 \to 0$	14906	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
7	$2 \to 1$	17093	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
8	$2 \to 0$	6158	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
8	$2 \to 1$	12719	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00

Table S7: Adjusting versus fixed widths of the Hold range at $f = 0.175$ over 50 starting markets. Probability brackets are 95% Wilson confidence intervals (CI). Sell is the mean fraction of active edge traders in Sell, averaged over steps after the rule change and starting markets.

h	Adaptive p [95% CI]	Frozen p [95% CI]	Adaptive Sell	Frozen Sell
0	0.34 [0.22, 0.48]	0.34 [0.22, 0.48]	0.000	0.000
0.1	0.66 [0.52, 0.78]	0.50 [0.37, 0.63]	0.154	0.038
0.5	0.56 [0.42, 0.69]	0.30 [0.19, 0.44]	0.158	0.045
1	0.48 [0.35, 0.61]	0.42 [0.29, 0.56]	0.142	0.058
2	0.30 [0.19, 0.44]	0.56 [0.42, 0.69]	0.096	0.107
5	0.30 [0.19, 0.44]	0.52 [0.39, 0.65]	0.136	0.271
10	0.28 [0.17, 0.42]	0.52 [0.39, 0.65]	0.113	0.189
∞	0.28 [0.17, 0.42]	0.52 [0.39, 0.65]	0.113	0.189

Table S8: Changes in how strongly trades affected price at $f = 0.20$, with edge traders responding to price, 50 seeds per cell. Drawdown is a percentage; rejection is the fraction of post-switch eligible orders rejected.

h	λ	Transition p	Median drawdown (%)	Median rejection
0.1	0.005	1.00	0.10	0.507
0.1	0.020	1.00	0.40	0.506
0.1	0.050	1.00	0.99	0.520
1	0.005	1.00	1.00	0.441
1	0.020	1.00	3.92	0.412
1	0.050	1.00	9.52	0.395
∞	0.005	1.00	7.10	0.000
∞	0.020	1.00	25.26	0.000
∞	0.050	1.00	49.56	0.000

Table S9: Changes in observation time and the required Sell fraction, with edge traders kept in Hold, over 50 starting markets. Entries are interpolated f_{50} [95% seed-bootstrap interval] for each crossing criterion. The horizon grids are independent of adaptive refinement, so their interpolated values can differ slightly from Supplementary Table S3.

L	Post-switch steps	Criterion 0.4	Criterion 0.5	Criterion 0.6
32	750	0.236 [0.228, 0.238]	0.249 [0.245, 0.255]	0.266 [0.260, 0.270]
32	1,500	0.216 [0.210, 0.220]	0.230 [0.228, 0.234]	0.245 [0.238, 0.250]
32	3,000	0.207 [0.205, 0.210]	0.219 [0.210, 0.225]	0.229 [0.226, 0.232]
64	3,000	0.172 [0.169, 0.176]	0.178 [0.176, 0.180]	0.186 [0.180, 0.188]
64	9,000	0.165 [0.163, 0.167]	0.167 [0.165, 0.168]	0.171 [0.169, 0.175]
128	3,000	0.163 [0.162, 0.164]	0.167 [0.165, 0.168]	0.169 [0.168, 0.171]
128	6,000	0.154 [0.153, 0.156]	0.157 [0.156, 0.158]	0.159 [0.158, 0.160]
128	12,000	0.150 [0.148, 0.151]	0.152 [0.150, 0.153]	0.153 [0.152, 0.155]

Table S10: Comparison of logistic fits with interpolation between tested fractions. Parentheses give seed-bootstrap standard errors (SE), not binomial standard errors treating fractions as independent. The last column is the location of the empirical susceptibility maximum: it is a sampled fraction with local grid resolution, not an interpolated optimum.

Boundary	h	L	Logistic f_{50} (SE)	Logistic width (SE)	f at peak
Endog.	1	32	0.212 (0.0016)	0.020 (0.0014)	0.217
Endog.	1	48	0.184 (0.0011)	0.014 (0.0011)	0.190
Endog.	1	64	0.175 (0.0007)	0.010 (0.0008)	0.182
Endog.	1	96	0.166 (0.0007)	0.008 (0.0006)	0.170
Endog.	1	128	0.165 (0.0006)	0.006 (0.0006)	0.172
Endog.	∞	32	0.212 (0.0015)	0.019 (0.0014)	0.225
Endog.	∞	48	0.186 (0.0013)	0.014 (0.0011)	0.190
Endog.	∞	64	0.178 (0.0007)	0.009 (0.0008)	0.185
Endog.	∞	96	0.168 (0.0007)	0.007 (0.0005)	0.170
Endog.	∞	128	0.166 (0.0004)	0.005 (0.0003)	0.168
Held	∞	32	0.218 (0.0016)	0.018 (0.0014)	0.225
Held	∞	48	0.189 (0.0011)	0.013 (0.0009)	0.190
Held	∞	64	0.178 (0.0010)	0.011 (0.0008)	0.185
Held	∞	96	0.169 (0.0006)	0.007 (0.0006)	0.172
Held	∞	128	0.166 (0.0005)	0.006 (0.0004)	0.168

Figures

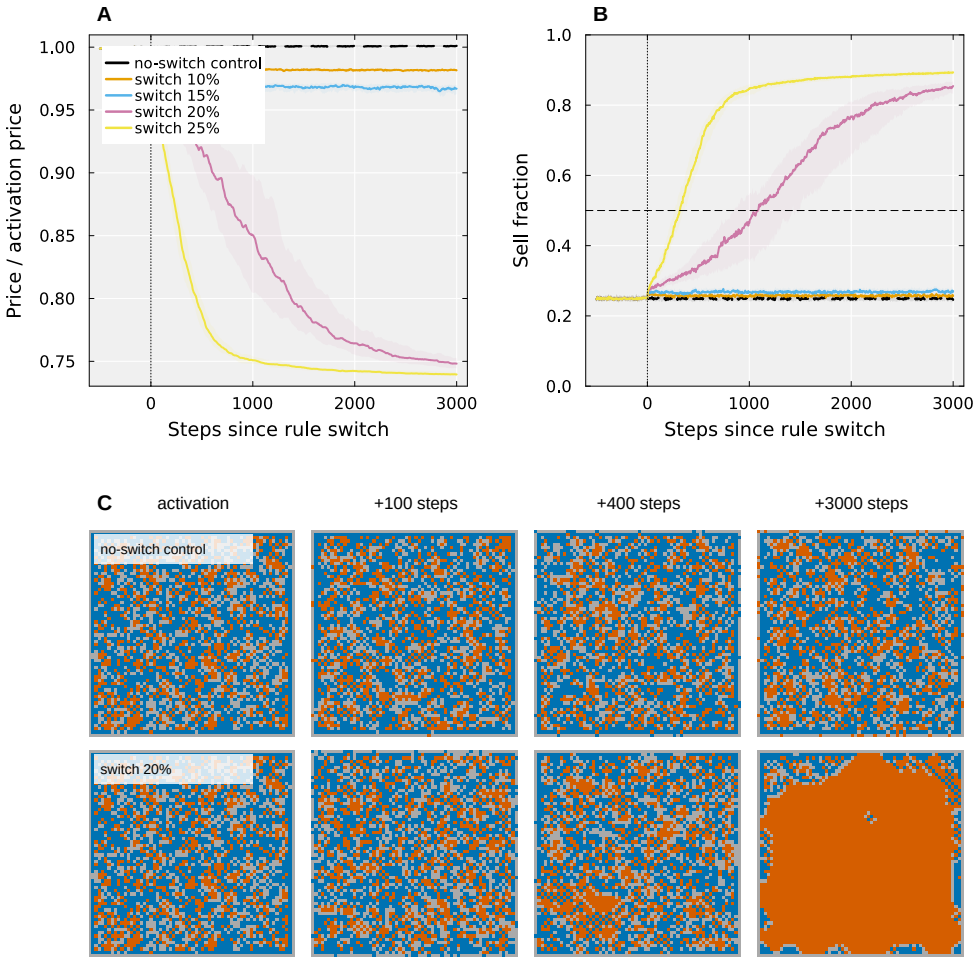


Figure 1: Changing the decision rule caused initially identical markets to develop differently. Fifty starting markets were simulated on a 64×64 grid for 1,000 steps before the rule change and 3,000 steps afterward. Each comparison market kept the original rule at the same selected sites. **(A)** Median price divided by its value when the rule changed; bands mark the interquartile range. **(B)** Median fraction of interior traders in Sell; the horizontal line marks one-half. **(C)** Snapshots of decisions for seed 12, the $f = 0.20$ pair nearest the median difference in peak Sell fraction between changed and unchanged markets. Blue, gray, and orange cells denote Buy, Hold, and Sell. Rows follow the two markets from their common state when the rule was assigned through 3,000 steps. All paired markets were identical before the vertical line marking the rule change.

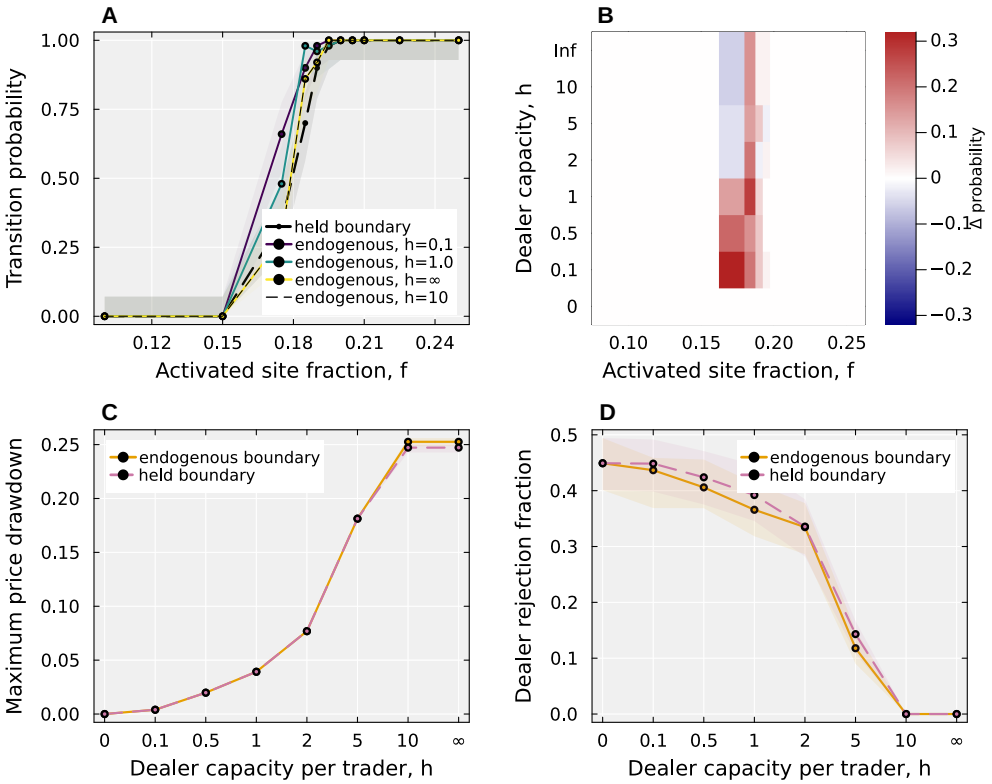


Figure 2: Changing decision rules while varying how much the dealer could buy and sell. **(A)** Probability of a Sell majority with edge traders kept in Hold and with those traders responding to price at selected dealer capacities. These conditions are labeled held boundary and endogenous boundary, respectively. Bands are 95% Wilson intervals over 50 starting markets. **(B)** Difference in majority probability between the condition with edge traders responding to price and the condition with them kept in Hold, across all capacities. **(C)** Median maximum drawdown at $f = 0.20$; bands mark the interquartile range. **(D)** Median fraction of orders backed by trader cash or shares that the dealer rejected at $f = 0.20$. Capacity $h = 10$ appears as a dashed overlay in (A) and a separate capacity in (B–D); it matched $h = \infty$ exactly.

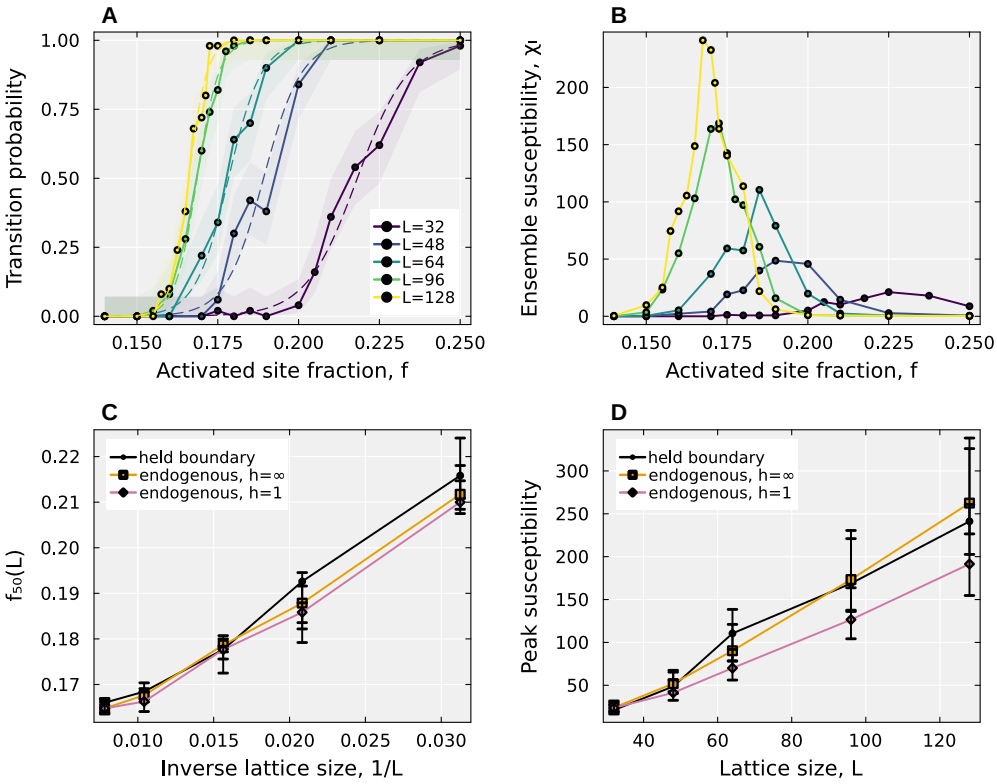


Figure 3: Effect of market size on the formation of a Sell majority over 50 starting markets. **(A)** Majority probability with edge traders kept in Hold; bands are 95% Wilson intervals. **(B)** Variation between runs with edge traders kept in Hold, measured as susceptibility: the variance of the final 500-step mean Sell fraction multiplied by the number of interior traders. **(C)** Fraction $f_{50}(L)$ at which half the runs reached a majority, estimated by interpolation for edge traders kept in Hold and for two dealer capacities with edge traders responding to price. **(D)** Maximum susceptibility versus grid size on linear axes. Error bars in (C,D) are 95% intervals from resampling starting markets. Solid lines connect tested values; dashed curves in (A) are logistic fits. No law for the limit of indefinitely large grids was fitted.

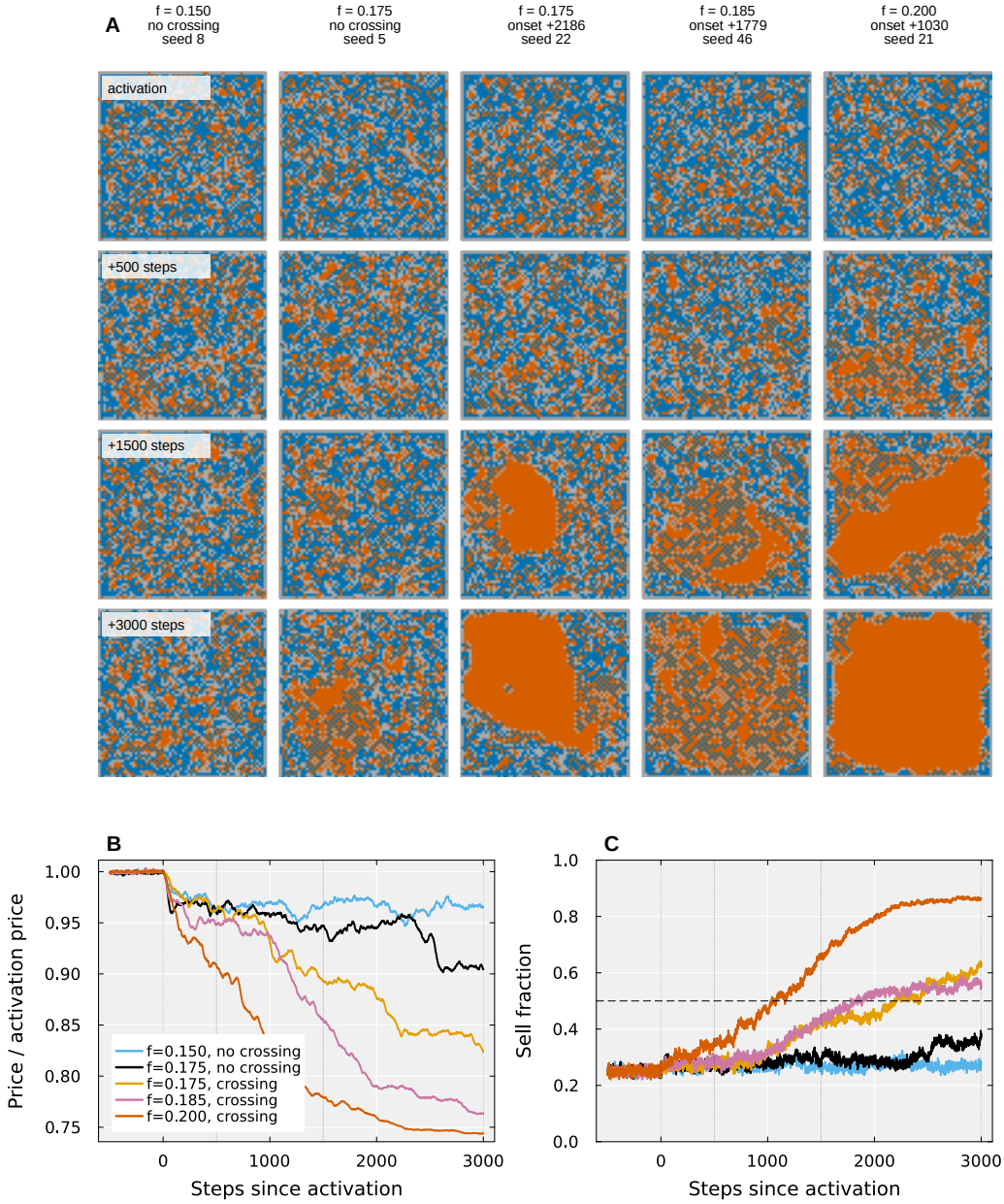


Figure 4: How traders' decisions changed over time when different fractions received the new rule, with edge traders responding to price and an unlimited dealer. Blue, gray, and orange cells denote Buy, Hold, and Sell. **(A)** Decisions in five runs (columns) when the rule changed and after 500, 1,500, and 3,000 steps (rows). From left to right, the runs include a case without a majority at $f = 0.15$, both outcomes at $f = 0.175$, and cases with a majority at $f = 0.185$ and 0.20. Runs with a majority were selected by proximity to their group's median time to a majority. Runs without a majority were selected by proximity to their group's median final Sell fraction. Selection preceded inspection of images. Seeds were 8, 5, 22, 46, and 21 from left to right. **(B)** Price divided by its value when the rule changed. **(C)** Interior Sell fraction; the horizontal line marks one-half. Vertical guides mark the four snapshots.

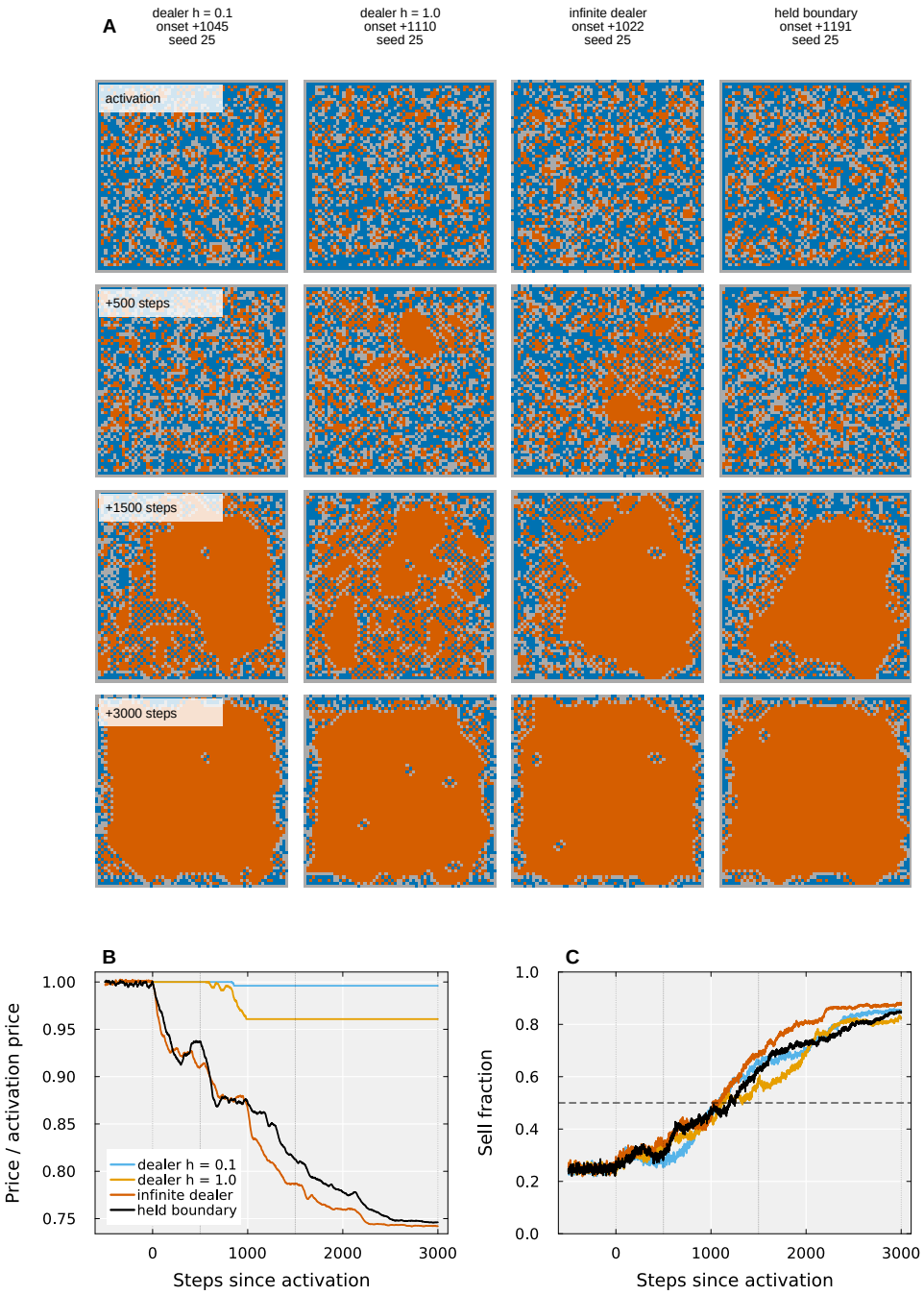


Figure 5: How the same starting market developed with different dealer capacities at $f = 0.20$. **(A)** Decisions under four conditions (columns) when the rule changed and after 500, 1,500, and 3,000 steps (rows). The first three columns allowed edge traders to respond to price, with dealer capacities $h = 0.1$, $h = 1$, and $h = \infty$. The last kept edge traders in Hold with $h = \infty$. Seed 25 was selected from runs that reached a majority at all three capacities by minimizing the sum of distances from each capacity's median time to a majority. Rows show the same elapsed times. **(B)** Price divided by its value when the rule changed. **(C)** Interior Sell fraction. Keeping edge traders in Hold prevented price from influencing decisions inside the grid; trades and price changes were still calculated. This example follows one starting market, while the full comparison measures outcomes across starting markets (Fig. 2).

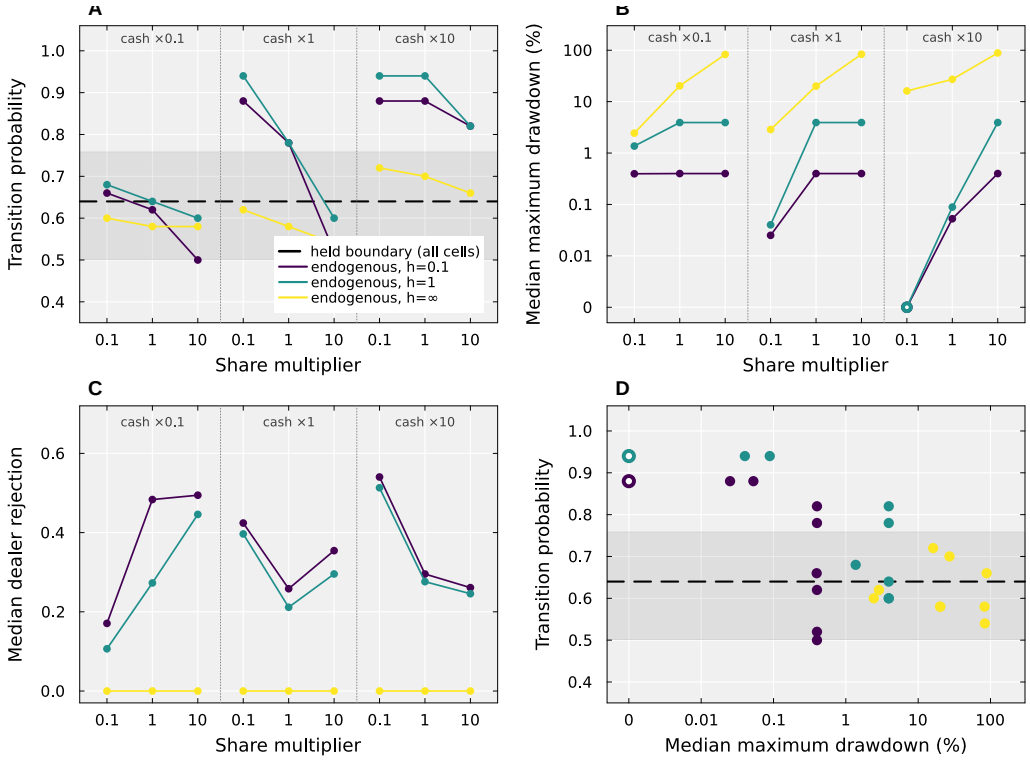


Figure 6: Effects of changing traders' cash and shares when the new decision rule was introduced, with edge traders responding to price, $f = 0.18$, and 50 starting markets. In A to C, the nine combinations are grouped by cash multiplier, with share multiplier on the axis and a separate series for each dealer capacity. **(A)** Probability of a Sell majority. The dashed line and band give the probability and its 95% Wilson interval with edge traders kept in Hold, which were identical in all nine combinations because resource changes could not affect decisions. **(B)** Median maximum drawdown on a logarithmic axis. Medians of exactly zero appear at the axis floor with open markers. **(C)** Median fraction of orders backed by trader cash or shares that the dealer rejected. **(D)** Majority probability against median maximum drawdown for all 27 combinations of cash, shares, and dealer capacity with edge traders responding to price. The axis and zero markers follow B, and the reference probability with edge traders kept in Hold is repeated.

A Supplementary methods and validation

A.1 One complete simulation step

Traders' current Buy, Hold, or Sell decisions generated orders before the new quote could influence their next decisions (Algorithm S1). We checked whether traders had the required cash or shares, matched orders, and completed trades before calculating a new price (Algorithm S2). Orders at step t came from the lattice X_{t-1} and stored boundary states $e_{i,t-1}$. Trades completed at the current price P_{t-1} before the difference between completed buys and sells generated P_t. Edge traders then used this price to decide whether to buy, hold, or sell, and their decisions influenced interior traders when calculating X_t. Rule activation occurred before this sequence without changing the current state. When we intentionally changed trader cash or shares, we did so before checking whether orders had the resources needed to complete. This timing explains why a rule switch leaves the first post-switch quote unchanged, whereas a financial intervention can affect it immediately. The comparison with edge traders kept in Hold used the same sequence but prevented these traders from responding to price. Under either boundary, a shortage of resources could stop an order without changing the trader's cellular-automaton state. We recorded orders excluded by trader resources as agent suppression and eligible orders left unfilled by the dealer as dealer rejection. Neighboring traders continued to respond to the stored Buy, Hold, or Sell state even when the associated order did not execute.

Algorithm S1 One market step with edge traders responding to price, $t-1 \to t$, in simulation order. Orders came from the old interior and boundary states, settled at the old quote, and generated the new quote. The new quote set the boundary before the interior cellular automaton update.

Require: lattice X_{t-1}; boundary states $e_i(t-1)$; trader books; dealer ledger; quote and band state $(P_{t-1}, \mu_{t-1}, \sigma_{t-1})$; rule field R_i; parameters; simulation seed
Ensure: lattice X_t; boundary states $e_i(t)$; settled books and dealer; quote and band state (P_t, μ_t, σ_t); all order counts

if t is the scheduled rule-switch step **then**
  $R_i \leftarrow 19286$ for every preselected target site i {preserve every dynamic state variable}
end if
if t is a scheduled balance-sheet intervention **then**
  for all traders i **do**
    $w_i \leftarrow g_w w_i$; $q_i \leftarrow \mathrm{round}(g_q q_i)$
  end for
end if
call Algorithm S2 using states at $t-1$ and P_{t-1}
receive $(n_t^B, n_t^S, u_t^B, u_t^S, r_t^B, r_t^S)$ and settled ledgers
$\phi_t \leftarrow (n_t^B - n_t^S)/N_{\mathrm{tr}}$
$P_t \leftarrow P_{t-1}\exp(\lambda\phi_t)$
$\mu_t \leftarrow \mu_{t-1} + \alpha(P_t - \mu_{t-1})$
$\sigma_t \leftarrow \sigma_{t-1} + \alpha(|P_t - \mu_t| - \sigma_{t-1})$
if boundary mode is endogenous **then**
  for all active boundary traders i **do**
    $e_i(t) \leftarrow 0$ if $P_t < \mu_t - k_i\sigma_t$; 2 if $P_t > \mu_t + k_i\sigma_t$; 1 otherwise
  end for
else
  $e_i(t) \leftarrow 1$ for every active boundary trader i
end if
write $e_i(t)$ to active boundary sites; leave passive sites at Hold
for all strictly interior sites i, simultaneously **do**
  $c_i(t) \leftarrow \sum_{j\in\partial i} s_j$ using old interior states and new boundary states
  $s_i(t) \leftarrow \mathcal{R}_{R_i}(c_i(t))$
end for

Algorithm S2 Order eligibility, dealer clearing, rationing, and settlement at P_{t-1}. The finite dealer filled only orders that other traders did not match. The $h = \infty$ branch filled every eligible order.

Require: old interior and boundary states; books $(w_i, q_i, \bar{c}_i)$; dealer (W_{t-1}, Q_{t-1}) and bounds $(Q_{\min}, Q_{\max}, C)$; P_{t-1}; h; cost-basis toggle; seed and step t

Ensure: settled books and dealer; executed counts (n_t^B, n_t^S); agent suppression (u_t^B, u_t^S); dealer rejection (r_t^B, r_t^S)

$\mathcal{B}, \mathcal{S} \leftarrow \varnothing$; $u_t^B, u_t^S \leftarrow 0$
for all interior traders, followed by active-boundary traders i **do**
  $o_i \leftarrow +1$ for Buy, -1 for Sell, and 0 for Hold {an edge trader reads its stored boundary state}
  if $o_i = +1$ and $w_i \geq P_{t-1}$ **then**
    append i to $\mathcal{B}$
  else if $o_i = -1$, $q_i \geq 1$, and the cost-basis test passes **then**
    append i to $\mathcal{S}$
  else if $o_i = +1$ **then**
    $u_t^B \leftarrow u_t^B + 1$
  else if $o_i = -1$ **then**
    $u_t^S \leftarrow u_t^S + 1$
  end if
end for
$B_t \leftarrow |\mathcal{B}|$; $S_t \leftarrow |\mathcal{S}|$; $n_t^B \leftarrow B_t$; $n_t^S \leftarrow S_t$
if $h < \infty$ and $B_t > S_t$ **then**
  $K_t^B \leftarrow Q_{t-1} - Q_{\min}$ {shares that the dealer could sell}
  $n_t^B \leftarrow S_t + \min(B_t - S_t, K_t^B)$
else if $h < \infty$ and $S_t > B_t$ **then**
  $K_t^S \leftarrow Q_{\max} - Q_{t-1}$ {inventory space}
  $K_t^S \leftarrow \min\big(K_t^S, \lfloor (W_{t-1} + C)/P_{t-1} \rfloor\big)$ {cash limit; $C = 0$ in this study}
  $n_t^S \leftarrow B_t + \min(S_t - B_t, K_t^S)$
end if
$r_t^B \leftarrow B_t - n_t^B$; $r_t^S \leftarrow S_t - n_t^S$
if $r_t^B > 0$ **then**
  sort $\mathcal{B}$ by $z(\text{seed}, t, i, +1)$, then by trader identity
end if
if $r_t^S > 0$ **then**
  sort $\mathcal{S}$ by $z(\text{seed}, t, i, -1)$, then by trader identity
end if
for all the first n_t^B traders i in $\mathcal{B}$ **do**
  $\bar{c}_i \leftarrow (\bar{c}_i q_i + P_{t-1})/(q_i + 1)$; $q_i \leftarrow q_i + 1$; $w_i \leftarrow w_i - P_{t-1}$
end for
for all the first n_t^S traders i in $\mathcal{S}$ **do**
  $q_i \leftarrow q_i - 1$; $w_i \leftarrow w_i + P_{t-1}$ {$\bar{c}_i$ was unchanged on a sale}
end for
$W_t \leftarrow W_{t-1} + P_{t-1}(n_t^B - n_t^S)$
$Q_t \leftarrow Q_{t-1} + n_t^S - n_t^B$
return $(n_t^B, n_t^S, u_t^B, u_t^S, r_t^B, r_t^S)$ and settled ledgers

A.2 Rationing and paired reproducibility

When buy and sell orders backed by trader resources differed, the dealer limited the remaining trades if it lacked the resources to complete them all. Each eligible trader on that side received a 64-bit score from a SplitMix64 finalizer applied to $(\text{seed}, t, \text{trader identity}, \text{side})$. Orders were sorted by score, using trader identity to break ties reproducibly. The dependence on t prevented a trader from retaining the same priority across repeated orders. Paired paths used the same score for a given (t, i, side) even after their eligible sets diverged. Scoring did not consume draws from the generator used to initialize the lattice and target sites. For each pair, we checked the frame at T_{pre}, its preceding Sell history, the boundary coordinates and band multipliers, and the price history through the first post-switch quote. For cash and share comparisons with edge traders kept in Hold, we also required the full Sell history and final grid to match exactly across all nine changes to trader resources. Exact pre-switch

agreement ensured that a pair reached activation with the same history. Exact post-switch agreement across held-boundary financial interventions ensured that trader resources affected execution without entering the local rule update.

A.3 Conservation and limiting cases

Settlement conserved total cash and shares across traders and dealer when no intervention occurred:

$$\sum_i w_i(t) + W_t = \sum_i w_i(0) + W_0, \tag{10}$$

$$\sum_i q_i(t) + Q_t = \sum_i q_i(0) + Q_0. \tag{11}$$

The maximum absolute cash error across the production dealer-capacity and grid-size simulations was 2.53×10^{-6}; the share error was zero. In the cash and share comparisons, checks were applied separately before and after the intentional resource change. Its maximum cash error was 9.47×10^{-7}, again with zero share error. Three limiting cases also supplied regression checks. Unlimited capacity filled every eligible order and reproduced the earlier fill-all implementation. Zero capacity allowed only trades between traders, giving equal numbers of completed buys and sells and a fixed price. Interior decisions were then identical whether edge traders used the price or were kept in Hold. Capacity $h = 10$ never rejected an order in the dealer-capacity simulations and matched the unlimited path. That last equality applies to the stated endowments and horizon. A different sequence of buy and sell orders, larger trader holdings, or longer observation could cause this dealer to run out of cash or reach its limit on shares.

A.4 Simulation design and reproducibility

We used four main sets of simulations to measure the effects of rule adoption, dealer capacity, grid size, and trader resources, totaling 30,550 simulations (Supplementary Table S1). Each used 50 seeds and 3,000 post-activation steps as fixed campaign settings. Pilot and full modes recorded how the experiments were developed. The seed count and horizon were fixed campaign settings, without a prospective power calculation or independent preregistration of the horizon. With 50 seeds, a probability near one-half has a binomial standard error of about 0.07. The dealer-capacity simulations reused one no-switch control within each $(h, \text{boundary}, \text{seed})$ cell. Its comma-separated value (CSV) file therefore stored the control metrics with each treated summary row rather than as a separate trajectory for every fraction. Checkpointed campaigns reconstructed a fixed row order after threaded computation. A resumed campaign therefore retained the same control comparisons and row order as an uninterrupted run. Reusing the no-switch path avoided repeating an identical simulation for each activation fraction.

The selected paired-switch campaign used $L = 64$, an unlimited dealer, and an endogenous boundary at $f = 0.10, 0.15, 0.20, 0.25$. The dealer-capacity comparison used the same lattice size at fractions 0.10, 0.15, 0.175, 0.185, 0.190, 0.195, 0.200, 0.205, 0.210, 0.225, and 0.250. It crossed these fractions with capacities $h = 0, 0.1, 0.5, 1, 2, 5, 10, \infty$ under both boundaries. The original finite-size grid used fractions 0.140, 0.150, 0.160, 0.170, 0.175, 0.180, 0.185, 0.190, 0.200, 0.210, 0.225, and 0.250 at $L = 32, 48, 64, 96, 128$. Its three conditions were a held boundary with $h = \infty$ and endogenous boundaries with $h = 1$ or ∞. The cash–share campaign crossed nine multiplier pairs with $f = 0.170, 0.175, 0.180, 0.185$ and $h = 0.1, 1, \infty$ under both boundaries. Capacity was defined per trader throughout; a fixed-absolute-capacity size series was not run. The supplementary controls below retain this initialization while varying the stated intervention, parameter, or observation window.

Reproducing a simulated market required the same random-number seed for the starting conditions and the same order of calculations. We ran the simulations in Julia 1.12 with the environment pinned by the repository manifest. Original campaigns used 12 simulation threads and one garbage-collection thread; supplementary campaigns used four or eight simulation threads and one garbage-collection thread. A seeded Xoshiro generator [47] supplied the setup draws. Finite-size and balance-sheet campaigns wrote atomic checkpoints, then aggregated them in a deterministic order. Tests covered parameter validation, dealer limits, conservation, delayed activation, boundary controls, and time-varying priorities. Stepwise comparisons checked the optimized lattice kernel against the general totalistic solver under both homogeneous and heterogeneous rule fields.

Bootstrap analysis and generated LaTeX tables used Python, with package versions recorded in `code/revision/requirements.txt`. Source code, environment manifests, derived data, and figure scripts are available at https://github.com/varnerlab/Wolfram-markets-abm-paper. The repository also retains the earlier external-price classroom implementation and its archived Invesco QQQ exchange-traded fund input for provenance. No result from that model entered the principal analysis or figures.

Majority probability measured how often at least half the interior traders decided to sell, and onset measured how long that took. We used these summaries to compare activation fractions, lattice sizes, and trader resources. In the selected switch campaign, transition status was evaluated from peak Sell fraction because the stored onset used zero for a path with no crossing. Median onset was therefore undefined at the two lower fractions, where no switched seed transitioned (Supplementary Table S2). The refined size comparisons used midpoints, widths, and susceptibility maxima calculated over each sampled fraction grid (Supplementary Table S3). The susceptibility peak location remained a sampled fraction rather than an interpolated optimum (Supplementary Table S10). In the financial campaign, all 27 held-boundary conditions returned one transition probability at each fraction. The 27 endogenous conditions spanned a range of probabilities (Supplementary Table S4). When edge traders stayed in Hold, majority probability depended on neighbors' decisions at the given fraction of rule adoption. Allowing edge traders to respond to price added the influence of price changes on later decisions. Each probability counted crossings within its stated observation window, and each interpolated midpoint depended on the fractions sampled around that crossing.

A.5 Additional controls and sensitivity analyses

We compared a one-time decision change with a lasting change in the rule used for later decisions. We also varied the response to price and the observation window to identify which parts of the model supported widespread selling. These controls were added in response to review and retained the original initialization and 1,000-step baseline. Unless specified otherwise, they used $L = 64$, $\lambda = 0.02$, $h = \infty$, and 3,000 post-intervention steps. Each simulation with a new rule or a one-time instruction to sell was paired with a simulation that kept the original rule. We required exact pre-intervention agreement of their lattice and Sell histories, and checked the price prehistory for financial arms. Comparisons that kept edge traders in Hold and measured only decisions used a faster implementation that updated decisions without calculating trades or prices, using the same starting markets and decision rules. Tests matched its full Sell history and final grid to the market simulator, and 1,800 overlapping archived seed/fraction outcomes also agreed. This implementation reproduced decisions when prices could not influence traders, but did not calculate financial outcomes.

A one-time decision to sell and alternative decision rules. We set the selected traders to Sell once while keeping their rule for later decisions unchanged (Supplementary Table S5). At activation, we wrote Sell at the rule-switch target sites before orders executed and retained Rule 19280 everywhere. This one-time change could therefore affect the first price after the intervention, whereas assigning a new rule preserved that price and changed later decisions. Fractions matched the number of targeted sites; sites already in Sell did not change state. For $f = 0.15, 0.20, 0.25$ under both boundaries, no pulse reached a majority in 50 seeds (0/50; Wilson upper bound 0.071). Persistent Rule 19286 activation yielded 0/50, 50/50, and 50/50, respectively. The Wilson lower bound for 50/50 was 0.929. Directing these traders to sell once therefore did not reproduce the majorities caused by a lasting change in their decision rule. This comparison applied to the selected traders and timing; changing decisions once in a different way could have different outcomes.

The rule screen tested all 18 ways to change one entry of Rule 19280 (Supplementary Table S6). Each of the nine neighbor sums had two alternative outputs. We activated each resulting rule at the same three fractions using 20 seeds under both boundaries. At $f = 0.20$, six rules produced majority crossings in all 20 seeds under each boundary, while the other twelve produced none. Effective changes included entries $c = 2, 3, 4, 5$ as well as the selected $c = 1$ change. Every change that replaced a Buy or Hold response with Sell at a neighbor sum from one to five produced a majority, as did the change from Buy to Hold at a sum of three; the same Buy-to-Hold change at a sum of one or two, every change that moved a response away from Sell, and every change at a sum of six or more produced none. The input being altered and its replacement output therefore mattered even when the number of changed entries was fixed. The screen was descriptive and used no adjustment for multiple comparisons. An observed 0/20 had a Wilson upper bound of 0.161, and 20/20 had a lower bound

of 0.839. The screen identified other rules capable of producing majorities over the tested window without defining a general rule for when a majority would form.

Fixing the width of the price range in which edge traders chose Hold. The price range within which edge traders chose Hold changed as recent prices became more or less variable, allowing the same departure from the reference price to produce different decisions over time (Supplementary Table S7). We held the width of this range fixed at its value immediately before the rule change. Traders retained their k_j values and the moving center continued to update. The half-width $k_j\sigma_{1000}$ then stayed constant within a path, although we continued calculating recent price variation for comparison. Each run with a fixed width shared the same history as its comparison with an adjusting width, so the fixed values could differ across starting markets and capacities. At $f = 0.175$, the difference in majority probability between $h = 0.1$ and $h = \infty$ changed from 0.38 with adaptation to -0.02 with frozen widths. The difference between the adjusting-width and fixed-width comparisons was 0.40, with a 95% seed-bootstrap interval of [0.16, 0.64]. Removing width adaptation therefore removed the low-capacity advantage in the point estimates. The fraction of edge traders in Sell also changed. Even with the width fixed, increasing capacity did not consistently raise or lower majority probability, and individual probability intervals were wide. Adjusting the width contributed to the smaller dealer's advantage but did not explain every effect of capacity on majority probability.

Price-impact sensitivity. The price-impact coefficient determined how strongly the price changed when completed trader buys and sells differed (Supplementary Table S8). At $f = 0.20$, we varied λ over $\{0.005, 0.02, 0.05\}$ and capacity over $h \in \{0.1, 1, \infty\}$. A smaller coefficient reduced the immediate price movement from a given imbalance. When edge traders were active, the changed price also affected the ranges they used to decide whether to buy, hold, or sell. All 50 runs reached a Sell majority in every tested combination. Median drawdown increased with capacity at each coefficient, although its magnitude changed by a factor of seven to ten across the coefficient range. Dealer resources thus limited the price decline across a decade of impact choices, while widespread selling persisted at this activation fraction throughout the sweep. These uncalibrated coefficients describe model sensitivity; they do not convert the simulated drawdowns into estimates for an empirical market. Both the number of unmatched orders the dealer could fill and the price change assigned to those trades determined the size of the price decline.

Observation time and the fraction of traders required to sell. A longer observation window gave the altered local dynamics more time to reach the selected Sell fraction (Supplementary Table S9). Held-boundary checks used 50 seeds on $f = 0.140, 0.145, \ldots, 0.250$, extended to 0.300 at $L = 32$ to bracket the short-horizon response. We compared the original 3,000 steps with $T_{\text{post}} = 3000(L/64)^z$ for $z = 1, 2$ at the endpoint sizes. These windows were 1,500 and 750 steps at $L = 32$, and 6,000 and 12,000 at $L = 128$. Both scalings preserved the ordering of endpoint midpoints. At $L = 64$, increasing the horizon from 3,000 to 9,000 steps lowered the majority midpoint from about 0.178 to 0.167. Longer endpoint runs likewise continued to move the curve. Changing the crossing criterion to 0.4 or 0.6 retained the endpoint ordering and downward drift with time. At $L = 64$ and 3,000 steps, the three criteria gave midpoints of approximately 0.172, 0.178, and 0.186. The qualitative size response persisted, but the numerical midpoint depended on both the criterion and horizon. Neither these comparisons nor the final 500-step mean established equilibration at any size.

Baseline characterization and long no-switch paths. The pre-switch paths remained mixed, with similar mean Sell occupancy in the two windows preceding activation. Across 50 endogenous unlimited-dealer seeds, the mean interior Sell fraction was 0.2494 over steps 501–750 and 0.2503 over steps 751–1,000. In the latter window, the median within-seed return standard deviation was 0.0003, and the median within-seed 1st and 99th percentiles were -0.00068 and 0.00066. The mean zero-return fraction was 0.006. These summaries described the starting markets without showing that their statistical properties had stopped changing or matched an observed market. We extended the first ten production seeds under each boundary for 100,000 additional no-switch steps. None crossed a majority (0/10 per boundary; Wilson upper bound 0.278 for the ensemble event probability). The extension followed existing seeds and was not an independent replication. It showed that those paths remained below a majority over that longer interval, while leaving global stability and the distinction between temporary patterns and patterns that would persist indefinitely unresolved.

Fit and grid uncertainty. A monotone logistic curve supplied a second estimate of transition midpoint and width alongside empirical interpolation (Supplementary Table S10). Interpolation used the first upward crossing of each target probability and allowed local decreases in the sampled curve. The logistic fit imposed a monotone response as a working shape model. Both analyses used 2,000 bootstrap replicates that resampled whole seeds across fractions. Keeping each seed together preserved the dependence arising from shared initialization, so standard errors reflected seed variation without treating fractions as independent observations. Each susceptibility replicate reselected the maximum over the sampled grid. Its location remained a sampled fraction with finite local resolution. The resulting intervals measured uncertainty conditional on the refined grid and observation horizon. They did not account for discretization error, adaptive-grid selection uncertainty, or changes caused by extending the horizon. No susceptibility exponent was estimated from these finite-size summaries.